



TerraMind AI

CAMPUS SUSTAINABILITY INTELLIGENCE

Explainable intervention-planning copilot for campuses

Scenario estimates you can defend · decision traces you can audit · fully offline, no API keys.

12 interventions

5 ranking modes

16 evidence sources

no API keys required

Why campus sustainability plans stall

Black-box models

Reviewers cannot interrogate the assumptions a number is built on — so they cannot approve it.

“Forecasts” oversell

“Predicted savings” reads like a promise. When reality diverges, the whole plan loses credibility.

SDG claims overstate

Tags turn into measurements. A campus gets credit as if it already did the work.

Start over every year

New spreadsheet, new analyst, same opaque result. Nothing accumulates — no learning curve.

The AI explains. The engine calculates. The trace is accountable.

Every number in TerraMind AI is produced by registered, deterministic calculators with versioned factors — never by the language model.

Model

Deterministic engine: a factor registry, a 12-action catalog, energy/water/waste/carbon calculators and a portfolio optimizer.

every result carries formula · unit · factor version

Explain

The copilot retrieves evidence, explains reasoning and drafts prose. It never computes a number on its own.

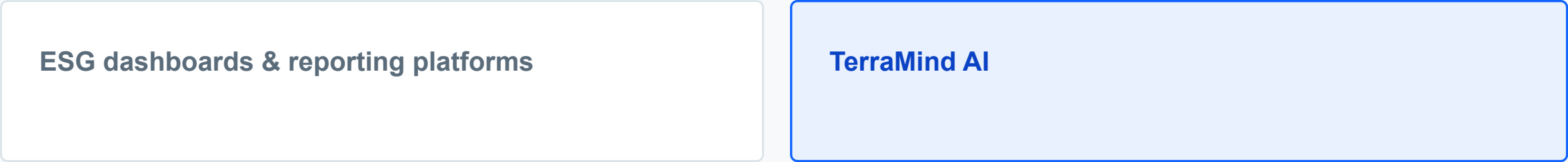
RAG over 16 evidence sources · citations on every claim

Decide

A decision trace records inputs, versions, weights, citations and approvals — a handover packet humans can sign.

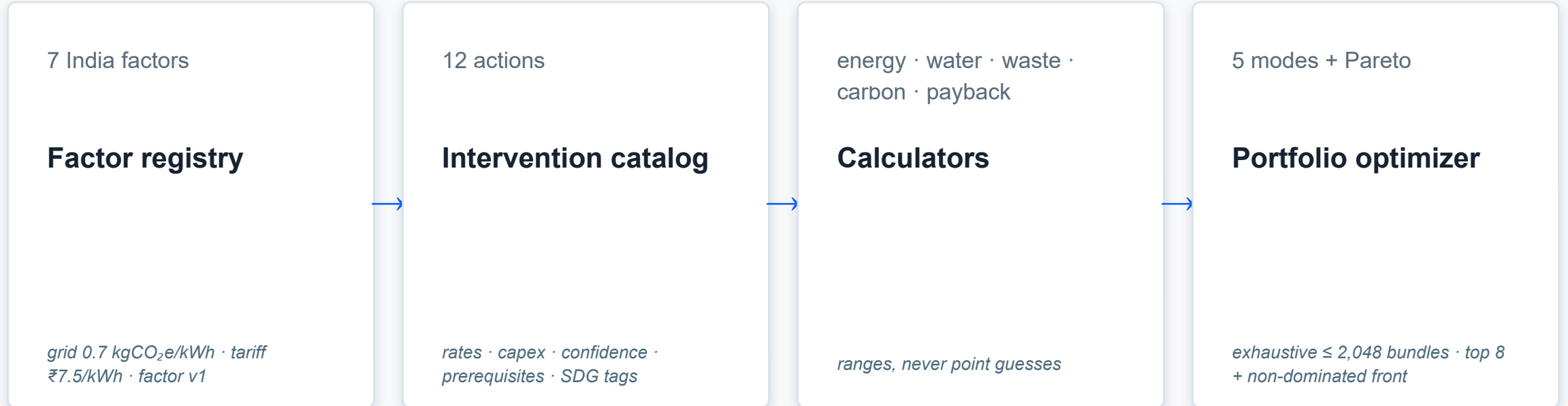
approval gate before adoption

The decision layer, not the dashboard



Primary question	How are we trending?	What should we do next?
Data posture	Warehouses the data you already have	Local baseline + explicit, versioned assumptions
Primary output	Reports and scorecards	A rank-ordered, costed action bundle
AI's role	Generates charts	Explains and reasons from the engine
What you walk away with	A score	A decision packet you can approve

Numbers with provenance



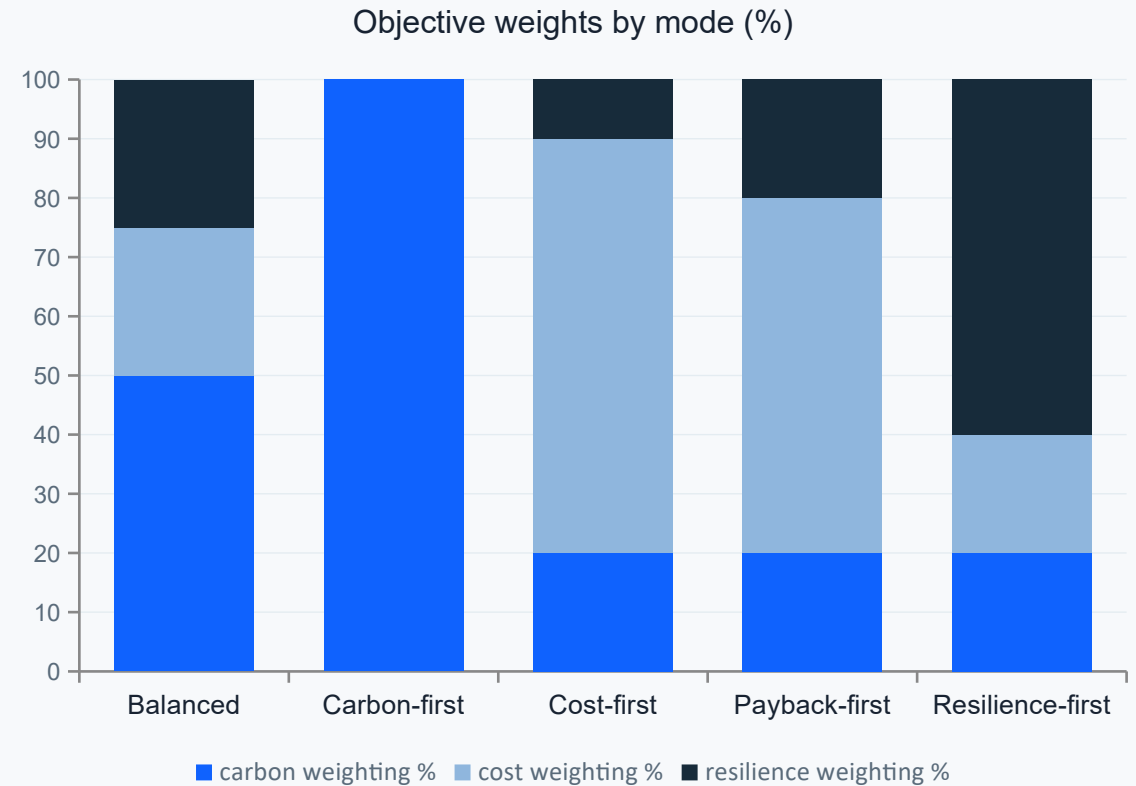
Uncertainty is carried, not hidden: every estimate is a low–high range whose width reflects data completeness, and the whole pipeline is reproducible from the recorded inputs.

Five modes, visible weights

Every ranking is a weighted sum of named objectives — and the weights are printed in the result.

No hidden 'recommended bundle': up to eight bundles are ranked on a transparent score, and the Pareto front shows alternatives a committee might prefer even when they are not top of the list.

balanced
carbon-first
cost-first
payback-first
resilience-first



Explains. Never invents.

Guardrails

Rejects forecasts, guarantees and 'certified savings' phrasing.



Grounding

Retrieves and cites evidence filtered by intervention and geography.



Reasoning

Calls the deterministic engine for every figure it shows.



Labelling

Labels every answer a 'scenario estimate' with its range.

Retrieved evidence

- “LED Lighting Fact Sheet” · US Department of Energy
- “Lighting Retrofit Best Practices” · BEE India
- “ISO 50001 Energy Management” · ISO

Citations are verifiable URLs — an AI can point to its sources, an engine never has to.

One decision, one auditable trace

- Every scenario run and every copilot answer writes a DecisionTrace.
- It records the input hash (SHA-256), model and factor versions, weights, citations and a full snapshot.
- Approvals and rejections are appended as their own trace events.
- Nothing a reviewer sees is unverifiable — reproduce the run, get the same numbers.

Inputs, versions and outcomes stay linked forever — your audit trail for the next committee.

DECISION TRACE · #49

kind	scenario run
input hash	7f3a...c9b2
model	campus-interventions-v1.0
factors	india-demo-factors-v1.0
weights	carbon .50 · cost .25 · res .25
citations	3 evidence sources
status	pending approval

Approve

Reject

A decision packet, not a dashboard

Decision packet · Campus LED + Solar

⚠ **scenario estimate — not a forecast**

- Executive summary
- Baseline & data quality
- Ranking & trade-offs
- Assumptions & factors
- SDG touchpoints
- Limitations & responsible-AI

- A single markdown document a committee can read in ten minutes — and approve in one.
- Contains only structured values produced by the engine — the AI writes prose, never numbers.
- Bundles the sequenced action plan, the ranges and the responsible-AI notes.
- PDF-ready today; the template is versioned for changes you can inspect.

gen → approval → adoption: the packet is the handover record, and it is traceable to every input.

Honest labels, honest ranges

Metric	Annual value	Label	Source
Electricity	100,000 kWh	entered	annual bill, FY 2025–26
Water	10,000 kL	entered	utility statement
Waste	20,000 kg	entered	waste audit estimate
Carbon	≈70 t CO ₂ e	derived	grid factor 0.7 kgCO ₂ e/kWh

- Five labels: measured · entered · derived · modeled · not-available.
- Missing data widens the range — it is never silently imputed.
- Measured and modeled values are never conflated.
- SDG tags are described as mappings, not measurements of impact.

Built to run where you run

pnpm monorepo

6 workspace packages · one-way
deps

TypeScript 5.9

strict · ESM · zero implicit any

tRPC v11 + superjson

typed end-to-end API · one endpoint

React 19 · Vite 7 · Tailwind 4

Carbon-inspired UI · light + dark

Drizzle ORM + MySQL 8

9 tables · JSON demo store

Deterministic core

ES2022 · pure functions · zod v4
inputs

JWT sessions (jose)

HttpOnly cookie · SameSite=Lax

Vitest · esbuild

41 tests · 66 KB API bundle

Runs fully offline · no API keys · demo store → production MySQL in one switch

One command from demo to production

Local demo server with a zero-setup
JSON store — instant onboarding.

pnpm dev

10 seconds to first run

Single Node process: SPA + typed API
from one port, ready for a VM.

pnpm build & start

one binary, 66 KB engine

API + MySQL 8 with healthchecks — the
same code, containerised.

docker compose up

drop-in for institutional IT

No external LLM, no cloud account, no data export — the demo is the product, just without the credentials.

From a single campus to an institutional portfolio

Near term

- Production auth provider
- Remote LLM toggle (Granite / OpenAI / Ollama)
- PDF decision packets
- Multi-site portfolio views

Medium term

- Postgres store
- Time-series baselines & seasonality
- Structured scenario diffing
- Shared-factor governance

Longer term

- On-field verification loop feeding the factor registry
- Community factor packs
- i18n on the evidence base

Turn numbers into decisions your committee can sign.

TerraMind AI — explainable intervention-planning copilot for campuses and institutional sites. Runs locally today; scales to institutional portfolios tomorrow.

MIT licensed

Offline demo included

Decision traces, first-class